Very similarly to the other orientation, we can split away objects of the type

$$1 \longrightarrow 0 \longleftarrow 0\ , \qquad 0 \longrightarrow 0 \longleftarrow 1\ ,$$

which results in a situation where both A and B are injective:

$$\underset{V}{\bullet} \xrightarrow{\ A\ } \underset{W}{\bullet} \xleftarrow{\ B\ } \underset{Y}{\bullet}\ .$$

By identifying V and Y as subspaces of W, this leads to the problem of classifying pairs of subspaces of a given space W up to isomorphism (the **pair of subspaces problem**). To do so, we first choose a complement W' of $V \cap Y$ in W and set $V' = W' \cap V$, $Y' = W' \cap Y$. Then we can decompose the representation as follows:

$$\begin{aligned} &\underset{V}{\bullet} \longrightarrow \underset{W}{\bullet} \longleftarrow \underset{Y}{\bullet} \\ &= \underset{V'}{\bullet} \longrightarrow \underset{W'}{\bullet} \longleftarrow \underset{Y'}{\bullet} \ \oplus\ \underset{V \cap Y}{\bullet} \xrightarrow{\sim} \underset{V \cap Y}{\bullet} \xleftarrow{\sim} \underset{V \cap Y}{\bullet}\ . \end{aligned}$$

The second summand is a multiple of the object $1 \xrightarrow{\sim} 1 \xleftarrow{\sim} 1$. We go on decomposing the first summand. Again, to simplify notation, we let

$$V = V', \qquad W = W', \qquad Y = Y'.$$

We can now assume that $V \cap Y = 0$. Next, let W' be a complement of $V \oplus Y$ in W. Then we get

$$\begin{aligned} &\underset{V}{\bullet} \longrightarrow \underset{W}{\bullet} \longleftarrow \underset{Y}{\bullet} \\ &= \underset{V}{\bullet} \longrightarrow \underset{V \oplus Y}{\bullet} \longleftarrow \underset{Y}{\bullet} \ \oplus\ \underset{0}{\bullet} \longrightarrow \underset{W'}{\bullet} \longleftarrow \underset{0}{\bullet}\ . \end{aligned}$$

The second of these summands is a multiple of the indecomposable object $0 \longrightarrow 1 \longleftarrow 0$. The first summand can be further decomposed as follows:

$$\begin{aligned} &\underset{V}{\bullet} \longrightarrow \underset{V \oplus Y}{\bullet} \longleftarrow \underset{Y}{\bullet} \\ &= \underset{V}{\bullet} \xrightarrow{\sim} \underset{V}{\bullet} \longleftarrow \underset{0}{\bullet} \ \oplus\ \underset{0}{\bullet} \longrightarrow \underset{Y}{\bullet} \xleftarrow{\sim} \underset{Y}{\bullet}\ . \end{aligned}$$

These summands are multiples of

$$1 \longrightarrow 1 \longleftarrow 0\ , \qquad 0 \longrightarrow 1 \longleftarrow 1$$